ModelIt!

Teacher's Guide

Can We Reverse Deforestation Before It's Too Late?

Modeling Carbon Sinks, Tipping Points, and Reforestation Economics

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-LS2-4, HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-4, HS-ESS3-3

Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem. Create a computational simulation of a designed solution for reducing the impacts of human activities on the environment and biodiversity.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical representations of carbon cycling rates, deforestation-reforestation timescale mismatches, and feedback loop dynamics to predict when forest systems cross tipping points.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy / ESS3.C Human Impacts on Earth Systems** — Students model how forests cycle carbon between biosphere and atmosphere, and how human land-use decisions disrupt these cycles with cascading effects on climate and regional hydrology.
Crosscutting Concept	**Stability and Change / Cause and Effect** — Students analyze how gradual deforestation can suddenly trigger cascading ecosystem collapse through nonlinear feedback loops that convert stable forests into unstable systems prone to savannification.

CAST Testing Alignment

MULTIPLE CHOICE:

The Amazon rainforest has lost approximately 17% of its original area. Research suggests the savannification tipping point occurs at 20-25% forest loss. Based on your model, what is the most accurate assessment of the current situation?

CONSTRUCTED RESPONSE:

Using your model, explain why planting one billion trees — while beneficial — cannot substitute for preventing deforestation of existing intact forests. Reference carbon stock timescales, evapotranspiration feedback, biodiversity, and tipping point dynamics in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Forests are Earth's most powerful terrestrial carbon sink, absorbing approximately 2.6 billion tons of CO_2 per year — about 30% of human emissions. But deforestation is rapidly converting this climate ally into a climate threat, releasing stored carbon while simultaneously dismantling the rainfall cycles that sustain the forest. The Amazon rainforest, which generates half its own rainfall through evapotranspiration, is approaching a tipping point that could trigger the irreversible collapse of the world's largest tropical forest.

Forests as Carbon Machines

A mature tropical forest stores approximately 250 tons of carbon per hectare in living biomass and 100-150 tons per hectare in soil organic matter. Globally, forests store approximately 860 billion tons of carbon — more than the entire atmosphere. Through photosynthesis, forests absorb CO_2 and lock it into wood, roots, and soil. When forest is cleared, this process reverses: burning releases carbon immediately as CO_2, while decomposition of roots and soil organic matter releases it over years to decades. Deforestation and forest degradation account for approximately 10-15% of global greenhouse gas emissions — more than all the world's cars combined.

The Evapotranspiration Feedback

Tropical forests don't just absorb rain — they create it. Through evapotranspiration, trees pump water from deep soil through their roots and release it as vapor through tiny pores in their leaves. This moisture rises, forms clouds, and falls as rain downwind. In the Amazon, up to 50% of rainfall is recycled water that has passed through tree canopies multiple times. This creates a self-sustaining system: the forest generates the rainfall it needs to survive. But when large areas are deforested, evapotranspiration declines, rainfall decreases, surviving forest dries out, fire risk increases, and more forest dies — a positive feedback loop toward savannification.

The Amazon Tipping Point

Research by Carlos Nobre and Thomas Lovejoy suggests that the Amazon rainforest faces an irreversible tipping point at approximately 20-25% total deforestation. At this threshold, the remaining forest can no longer generate sufficient rainfall to sustain itself. The transition is nonlinear: it doesn't happen gradually but rather as a sudden phase shift when the rainfall-fire feedback loop overwhelms the forest's resilience. As of 2025, approximately 17% of the Amazon has been deforested — dangerously close to the projected tipping point. In the southeastern Amazon, the dry season has already lengthened by 4-5 weeks since 1979, a clear sign that the rainfall cycle is weakening.

Reforestation: Promise and Limitations

Reforestation is widely promoted as a climate solution — the Bonn Challenge aims to restore 350 million hectares by 2030. However, several critical limitations constrain its effectiveness. First, timescale: a newly planted tree takes 20-40 years to reach carbon sequestration maturity, while the carbon from a cleared forest is released immediately. Second, quality: monoculture tree plantations sequester carbon but support a fraction of native biodiversity and cannot replicate the hydrological functions of natural forest. Third, permanence: planted forests are vulnerable to fire, drought, and future clearing — carbon credits based on tree planting can be reversed. Fourth, economics: the land cleared for agriculture generates immediate income, while reforestation requires sustained investment with returns decades away.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the forest carbon system: Deforestation Rate, Forest Carbon Stock, Evapotranspiration Rate, Regional Rainfall, Fire Susceptibility, and Reforestation Rate — distinguishing the two controllable human decisions (deforestation and reforestation) from internal ecosystem responses.

E **Establish Relationships**

Students determine that Deforestation reduces Forest Carbon Stock and Evapotranspiration Rate, which reduces Regional Rainfall, which increases Fire Susceptibility, which causes additional forest loss — a self-reinforcing loop. Reforestation counteracts this but operates on a much longer timescale.

V **Visualize & Model**

Students build a computational model showing how the deforestation-drought-fire feedback loop accelerates forest loss and carbon release, with a tipping point at which the loop becomes self-sustaining.

E **Evaluate Outcomes**

Students run current trajectory, reforestation matching, and tipping point scenarios to identify the forest loss threshold at which the feedback loop overwhelms reforestation efforts and triggers irreversible savannification.

R **Revise & Extend**

Students add Biodiversity Index, Soil Carbon Release, or Agricultural Commodity Price to explore how ecological complexity and economic drivers affect tipping point dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This line is the most dangerous border on Earth. On one side, a climate machine that's been running for millions of years. On the other side, cattle pasture. And the border is moving."*

Watch For

Show satellite time-lapse of Amazon deforestation from 2000-2025. The visual impact of forest disappearing in real time is powerful. Ask: How fast does it look like it's accelerating?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling a system that creates its own weather, stores twice as much carbon as your car will ever produce, and is about to cross a point of no return."*

Watch For

Have students read objectives. Pre-teach 'evapotranspiration' — demonstrate with a sealed plastic bag over a houseplant to show water vapor release. Explain how this scales to an entire rainforest.

Slide 3: Big Question

3 min

Say: *"Here's the terrifying math: the Amazon has lost 17% of its forest. Scientists say the tipping point is at 20-25%. We're running out of margin."*

Watch For

Quick calculation: If 17% is gone and 20% is the tipping point, how many more hectares can be lost? At current rates, how many years is that? Let students do the math.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of a system that sustains itself — until it can't. The key is finding where the self-sustaining mechanism breaks."*

Watch For

Preview each LEVER step. Emphasize the unique feature: this system has a positive feedback loop (forest creates rain creates forest) that works in BOTH directions — sustaining forest OR destroying it.

Slide 5: Activity 1: Components

8 min

Say: *"Two human decisions — cut trees or plant trees. Four ecosystem responses. The question is whether planting can ever catch up to cutting."*

Watch For

Guide sorting of external vs. internal components. Deforestation Rate and Reforestation Rate are external (human-controlled). Everything else responds to the system. Discuss the timescale asymmetry.

Slide 6: Activity 2: Connections

8 min

Say: *"Find the death spiral: cut forest, lose rainfall, gain fire, lose more forest, lose more rainfall, gain more fire. Sound familiar? It's the same pattern we saw with coral reefs."*

Watch For

Help students map the reinforcing loop. Compare to the coral reef death spiral. Identify the key difference: humans directly drive the initial loss (deforestation) and can directly counteract it (reforestation) — but can they act fast enough?

Slide 7: Activity 3: Simulation

10 min

Say: *"Three futures: keep cutting, start replanting, or push past the tipping point. Which ones can we come back from?"*

Watch For

Students predict which scenarios are reversible before running simulations. Focus on the tipping point scenario — identify exactly when savannification becomes irreversible. Compare the timescale of deforestation damage to reforestation recovery.

Slide 8: Discoveries

5 min

Say: *"Your model just proved that a billion new trees won't save us if we keep cutting the old ones. Why? Because old forest and new trees are not the same thing."*

Watch For

Lead discussion on why protecting intact forest is more important than replanting. A 200-year-old tree stores more carbon, generates more rainfall, and supports more biodiversity than a 5-year-old seedling. Discuss the concept of irreplaceable ecosystems.

Slide 9: STEM Challenge

12 min

Say: *"You've been hired by a tropical nation. 70% of their forest still stands, but ranchers are clearing it fast. Design a strategy that keeps the trees standing — and keeps the people prosperous."*

Watch For

Groups design integrated conservation-economics strategies. Must address enforcement, reforestation, economic alternatives, and carbon market mechanisms. Present as government policy proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Deforestation Rate, Reforestation Rate
- **INTERNAL:** Forest Carbon Stock, Evapotranspiration Rate, Regional Rainfall, Fire Susceptibility

Relationships

- **Deforestation Rate to Forest Carbon Stock:** NEGATIVE (-) — Higher Deforestation Rate directly reduces Forest Carbon Stock by removing trees that store carbon in biomass and exposing soil carbon to decomposition. A cleared hectare of tropical forest releases approximately 350-400 tons of CO2.
- **Forest Carbon Stock to Evapotranspiration Rate:** POSITIVE (+) — More forest means more evapotranspiration, as living trees pump water from soil and release it as vapor. Forest Carbon Stock serves as a proxy for canopy area — more stored carbon means more trees actively cycling water.
- **Evapotranspiration Rate to Regional Rainfall:** POSITIVE (+) — Higher evapotranspiration generates more atmospheric moisture, forming clouds that produce rainfall. In the Amazon, this moisture recycling provides up to 50% of total precipitation — making the forest essential to its own water supply.
- **Regional Rainfall to Fire Susceptibility:** NEGATIVE (-) — Higher rainfall keeps forests humid and resistant to fire. Lower rainfall dries out leaf litter and living vegetation, making forests increasingly flammable. The relationship is strongly nonlinear — small decreases in rainfall near the dry-season threshold cause large increases in fire risk.
- **Fire Susceptibility to Forest Carbon Stock:** NEGATIVE (-) — Fire destroys forest biomass, releasing carbon immediately and reducing Forest Carbon Stock. This closes the positive feedback loop: less forest means less rain means more fire means less forest — a self-reinforcing spiral toward savannification.

Key Discoveries

- Tropical forests are not just carbon storage — they are active climate machines that generate up to 50% of their own rainfall through evapotranspiration, creating a self-sustaining system that collapses when too much forest is removed
- The Amazon rainforest is approaching a critical tipping point: research suggests that losing 20-25% of the original forest could trigger a self-reinforcing drought-fire cycle leading to savannification of 40-60% of the remaining forest
- Reforestation works but operates on fundamentally different timescales than deforestation — a chainsaw can clear a hectare in hours, but a replanted forest takes 20-40 years to reach carbon sequestration maturity, creating a dangerous time gap
- Converting forest to agriculture is economically rational for individual landowners but catastrophically irrational at the system level — the ecosystem services provided by intact forest (rainfall generation, carbon storage, biodiversity) are worth far more than the land's agricultural value

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Planting trees can fully offset fossil fuel emissions"	While reforestation is valuable, the scale is insufficient to offset current fossil fuel emissions. Global forests absorb about 2.6 billion tons of CO_2 per year, but humans emit approximately 37 billion tons per year from fossil fuels alone. Even doubling global forest area — a physical impossibility — could not keep pace. Furthermore, tree carbon storage is impermanent — fire, drought, or future clearing can release it all back.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All tree planting is equally beneficial"	Not all reforestation is equal. Monoculture tree plantations (single species, even spacing) sequester carbon but support minimal biodiversity, provide weak hydrological services, and are vulnerable to disease. Natural forest restoration using diverse native species provides 40x more biodiversity, generates more rainfall, and creates more resilient carbon storage. Some tree planting projects have actually replaced natural grassland ecosystems, causing net ecological harm.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Deforestation only matters for climate because of carbon"	While carbon release is the most quantified impact, deforestation damages multiple interconnected systems simultaneously. It disrupts rainfall patterns affecting agriculture thousands of kilometers away, eliminates biodiversity that may include undiscovered medicines, destroys indigenous communities' homelands, increases flooding and erosion, and degrades soil that took thousands of years to form. The model shows that the rainfall feedback alone can trigger regional climate disruption far beyond the carbon impact.